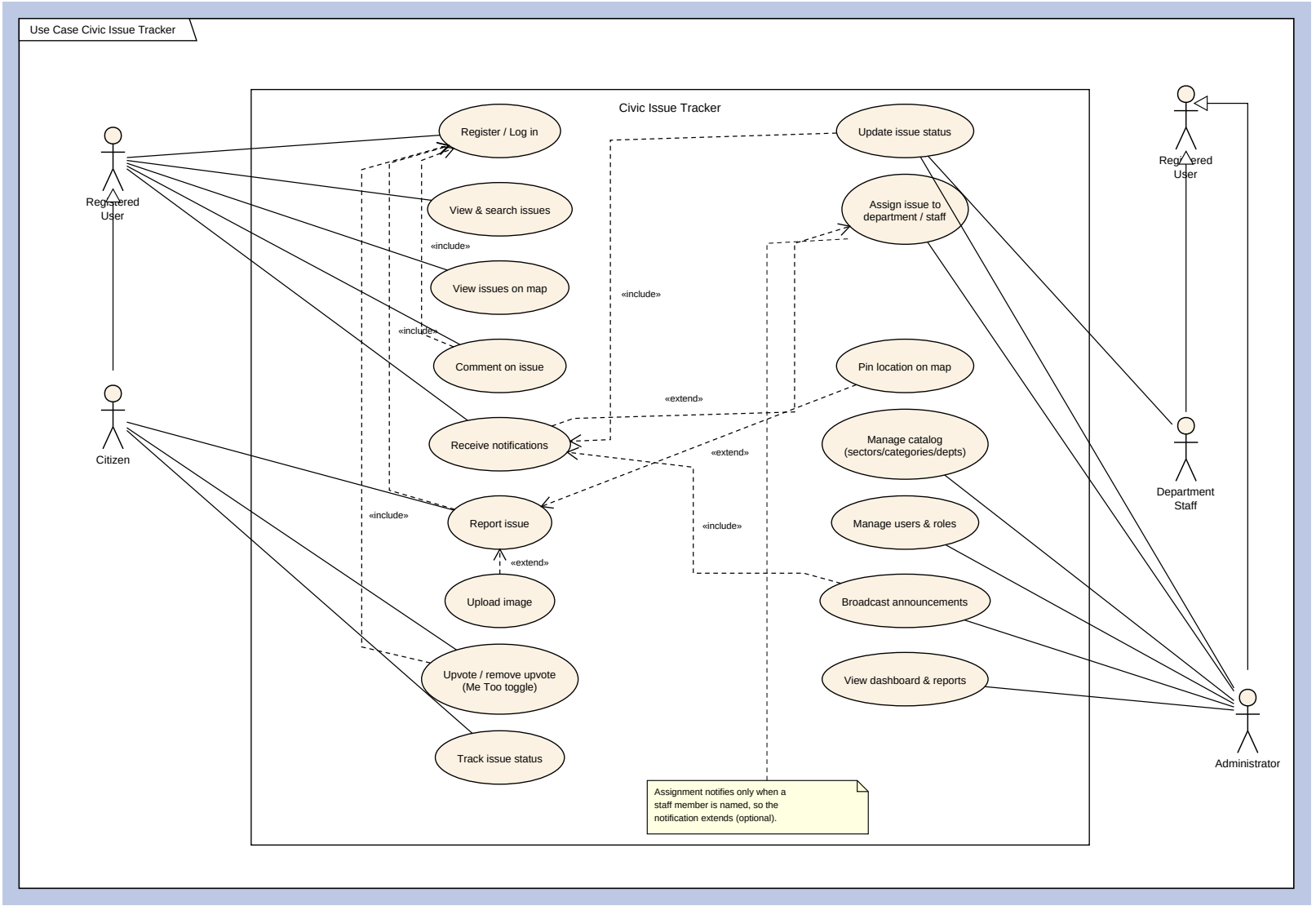
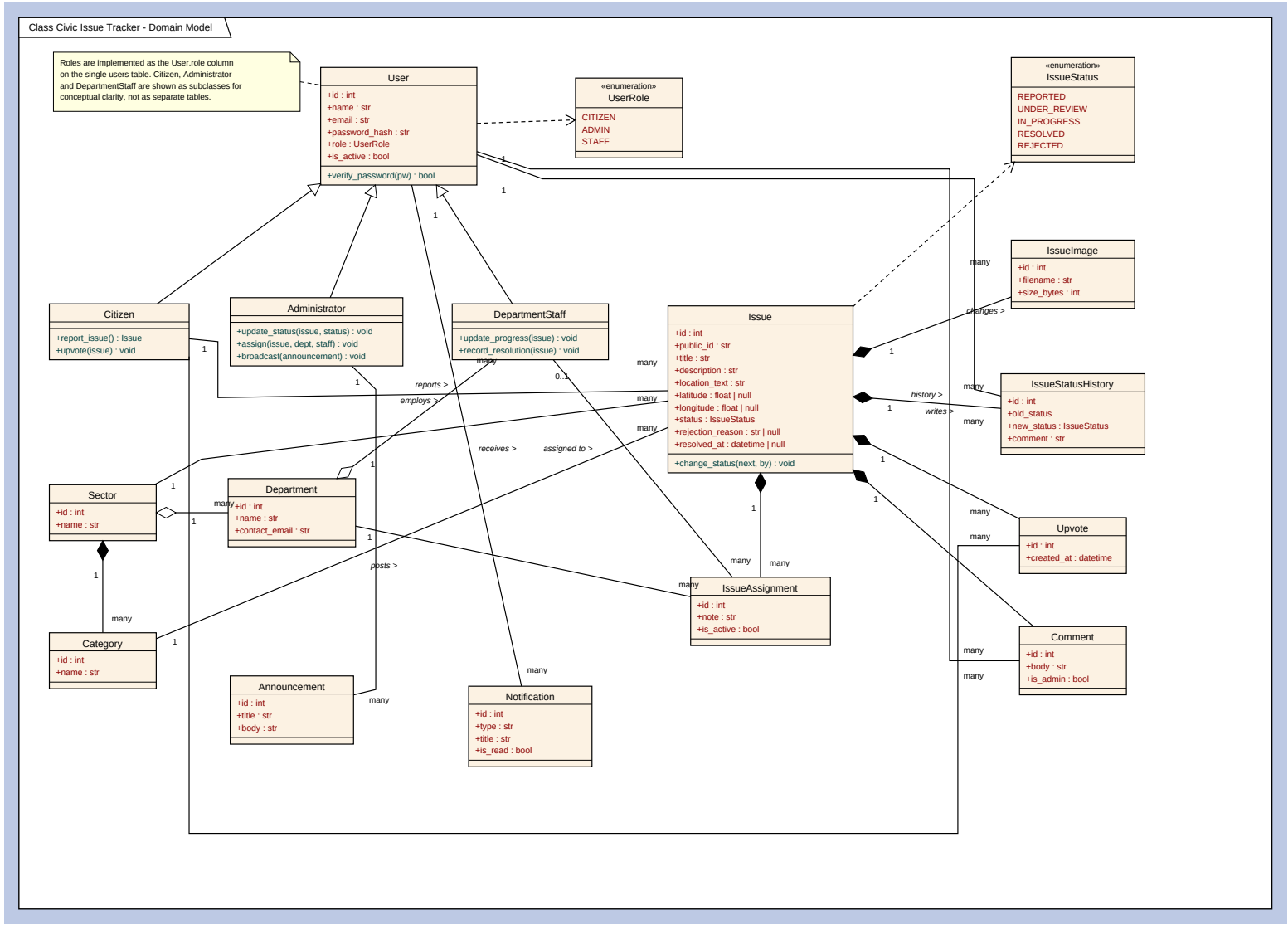
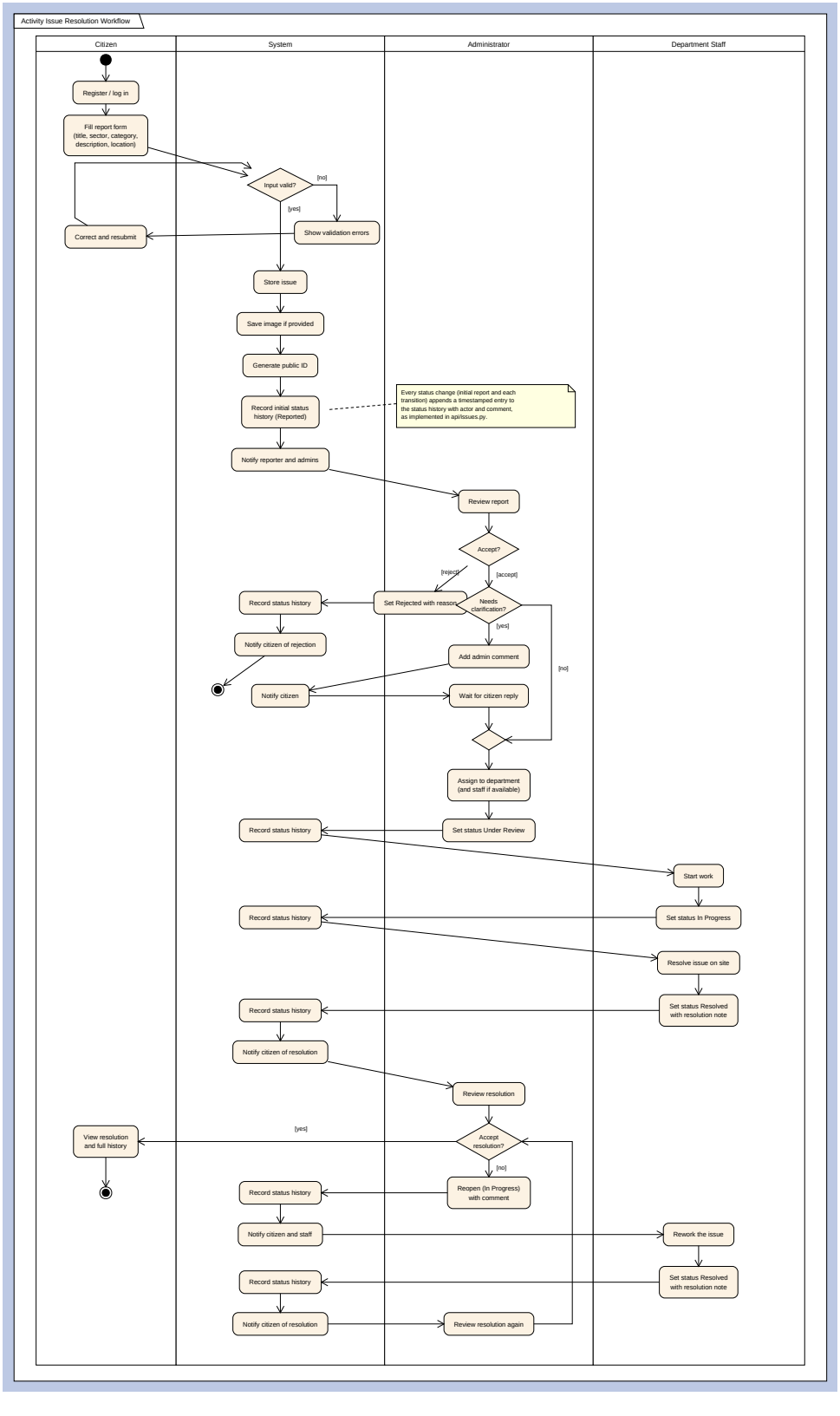
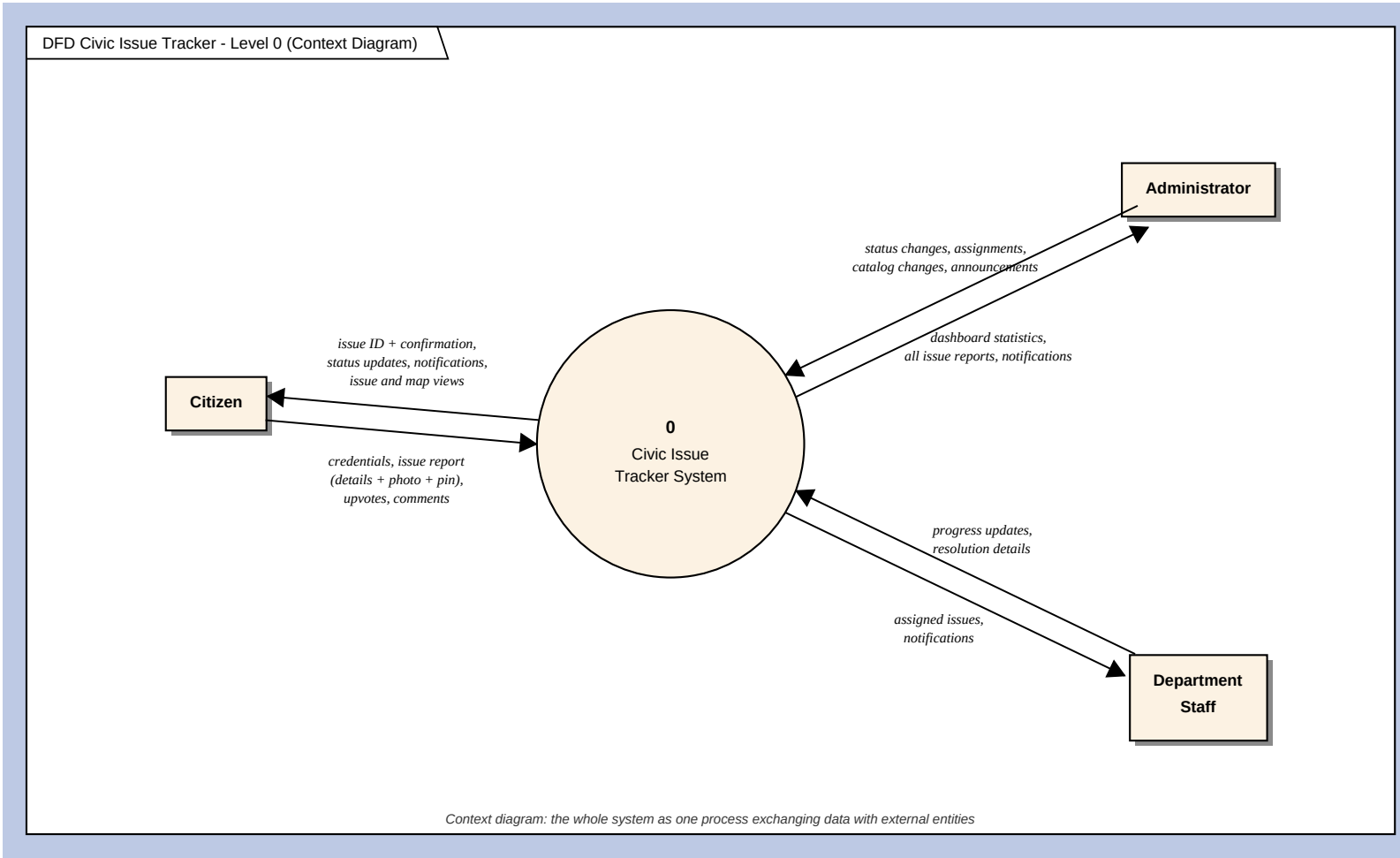
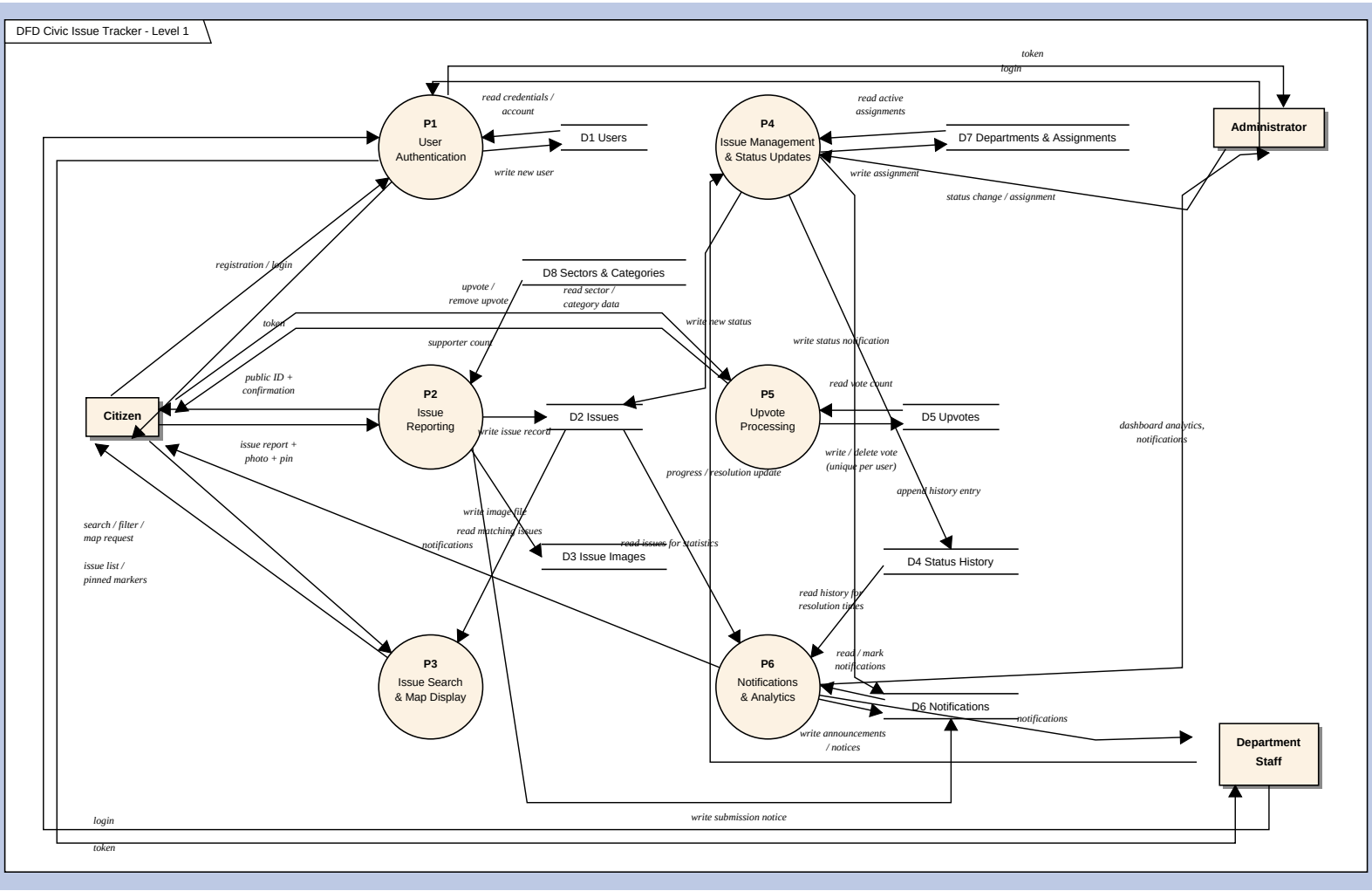
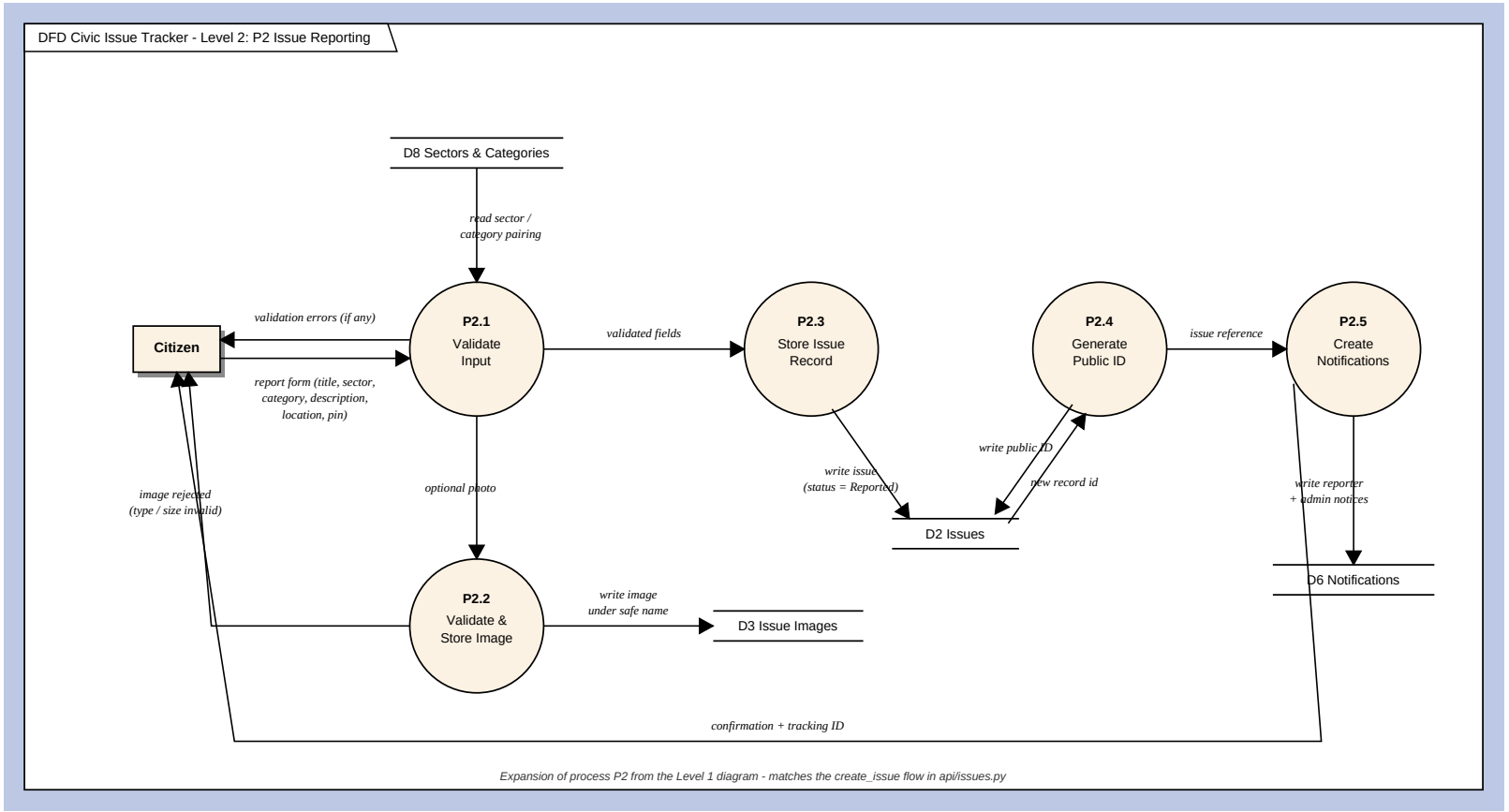

Use Case Civic Issue Tracker
Civic Issue Tracker
Registered User
Citizen
Registered User
Department Staff
Administrator
Register / Log in
View & search issues
View issues on map
Comment on issue
Receive notifications
Report issue
Upload image
Upvote / remove upvote (Me Too toggle)
Track issue status
Update issue status
Assign issue to department / staff
Pin location on map
Manage catalog (sectors/categories/depts)
Manage users & roles
Broadcast announcements
View dashboard & reports
«include»
«include»
«include»
«include»
«include»
«extend»
«extend»
«extend»
Assignment notifies only when a staff member is named, so the notification extends (optional).


Class Civic Issue Tracker - Domain Model
Roles are implemented as the User.role column on the single users table. Citizen, Administrator and DepartmentStaff are shown as subclasses for conceptual clarity, not as separate tables.
User
+id : int
+name : str
+email : str
+password_hash : str
+role : UserRole
+is_active : bool
+verify_password(pw) : bool
«enumeration»
UserRole
CITIZEN
ADMIN
STAFF
«enumeration»
IssueStatus
REPORTED
UNDER_REVIEW
IN_PROGRESS
RESOLVED
REJECTED
Citizen
+report_issue() : Issue
+upvote(issue) : void
Administrator
+update_status(issue, status) : void
+assign(issue, dept, staff) : void
+broadcast(announcement) : void
DepartmentStaff
+update_progress(issue) : void
+record_resolution(issue) : void
Issue
+id : int
+public_id : str
+title : str
+description : str
+location_text : str
+latitude : float | null
+longitude : float | null
+status : IssueStatus
+rejection_reason : str | null
+resolved_at : datetime | null
+change_status(next, by) : void
IssueImage
+id : int
+filename : str
+size_bytes : int
IssueStatusHistory
+id : int
+old_status
+new_status : IssueStatus
+comment : str
Sector
+id : int
+name : str
Department
+id : int
+name : str
+contact_email : str
Category
+id : int
+name : str
IssueAssignment
+id : int
+note : str
+is_active : bool
Upvote
+id : int
+created_at : datetime
Comment
+id : int
+body : str
+is_admin : bool
Announcement
+id : int
+title : str
+body : str
Notification
+id : int
+type : str
+title : str
+is_read : bool
reports >
employs >
receives >
assigned to >
posts >
changes >
history >
writes >
1
many
0..1


Activity Issue Resolution Workflow
Citizen
System
Administrator
Department Staff
Register / log in
Fill report form (title, sector, category, description, location)
Input valid?
[no]
[yes]
Show validation errors
Correct and resubmit
Store issue
Save image if provided
Generate public ID
Record initial status history (Reported)
Every status change (initial report and each transition) appends a timestamped entry to the status history with actor and comment, as implemented in api/issues.py.
Notify reporter and admins
Review report
Accept?
[reject]
[accept]
Set Rejected with reason
Record status history
Notify citizen of rejection
Needs clarification?
[yes]
[no]
Add admin comment
Wait for citizen reply
Notify citizen
Assign to department (and staff if available)
Set status Under Review
Record status history
Start work
Set status In Progress
Record status history
Resolve issue on site
Set status Resolved with resolution note
Record status history
Notify citizen of resolution
Review resolution
Accept resolution?
[yes]
[no]
View resolution and full history
Reopen (In Progress) with comment
Record status history
Notify citizen and staff
Rework the issue
Set status Resolved with resolution note
Record status history
Notify citizen of resolution
Review resolution again


DFD Civic Issue Tracker - Level 0 (Context Diagram)
Administrator
status changes, assignments,
catalog changes, announcements
dashboard statistics,
all issue reports, notifications
issue ID + confirmation,
status updates, notifications,
issue and map views
Citizen
credentials, issue report
(details + photo + pin),
upvotes, comments
0
Civic Issue
Tracker System
progress updates,
resolution details
assigned issues,
notifications
Department
Staff
Context diagram: the whole system as one process exchanging data with external entities


DFD Civic Issue Tracker - Level 1
token
login
read credentials / account
P1
User Authentication
D1 Users
write new user
read active assignments
P4
Issue Management & Status Updates
D7 Departments & Assignments
write assignment
status change / assignment
Administrator
registration / login
D8 Sectors & Categories
upvote / remove upvote
read sector / category data
token
write new status
supporter count
write status notification
public ID + confirmation
Citizen
P2
Issue Reporting
D2 Issues
write issue record
P5
Upvote Processing
read vote count
D5 Upvotes
dashboard analytics, notifications
issue report + photo + pin
write / delete vote (unique per user)
progress / resolution update
append history entry
write image file
read matching issues
notifications
read issues for statistics
D3 Issue Images
D4 Status History
search / filter / map request
issue list / pinned markers
read history for resolution times
read / mark notifications
P3
Issue Search & Map Display
P6
Notifications & Analytics
D6 Notifications
notifications
write announcements / notices
Department Staff
write submission notice
login
token


DFD Civic Issue Tracker - Level 2: P2 Issue Reporting
D8 Sectors & Categories
read sector / category pairing
Citizen
validation errors (if any)
report form (title, sector, category, description, location, pin)
P2.1 Validate Input
validated fields
P2.3 Store Issue Record
P2.4 Generate Public ID
issue reference
P2.5 Create Notifications
write public ID
new record id
write issue (status = Reported)
D2 Issues
write reporter + admin notices
D6 Notifications
image rejected (type / size invalid)
optional photo
P2.2 Validate & Store Image
write image under safe name
D3 Issue Images
confirmation + tracking ID
Expansion of process P2 from the Level 1 diagram - matches the create_issue flow in api/issues.py